

Arun Rimal

aarunrimal92@gmail.com — linkedin.com/in/arun-rimal — github.com/arunrimal — arunrimal.github.io

SUMMARY

Data & ML Engineer with 9+ years of professional experience spanning data engineering, ETL pipeline development, and applied machine learning. Currently pursuing an MS in Data Science at Wichita State University, with hands-on expertise in cloud platforms (AWS, Azure, Snowflake), LLM frameworks (LangChain, LlamaIndex), and full-stack AI application development. MBA background enables effective collaboration with business stakeholders to translate requirements into scalable data solutions.

PROFESSIONAL EXPERIENCE

Graduate Research Assistant

Jan 2025 – Present

National Institute for Aviation Research (NIAR), Wichita State University, Wichita, KS

- Support reverse engineering and redesign initiatives for aviation components using digital modeling data and Python-based analysis tools.
- Contribute to system enhancements through data processing workflows and technical documentation aligned with aerospace engineering standards.

Software Engineer, Data

Feb 2024 – Jul 2024

Infinite Computer Solutions, Kathmandu, Nepal

- Developed and maintained ETL pipelines and stored procedures using MSSQL and SSIS to process US healthcare data across multiple client environments.
- Improved pipeline reliability by refactoring legacy procedures, reducing manual intervention in data transformation workflows.

Data Engineer

Sep 2021 – Feb 2024

Devfinity Nepal, Kathmandu, Nepal

- Built scalable ETL pipelines in Python and SQL for API ingestion and database replication across multiple production environments.
- Designed and deployed end-to-end data warehouse pipelines in Azure Data Factory (Transact → Staging → Production), improving data freshness for downstream reporting.
- Automated web data ingestion using Python (Selenium) and integrated with Azure Blob Storage, eliminating manual data collection workflows.
- Processed high-volume JSON/XML data using concurrent processing techniques, improving throughput and reducing job execution time.
- Developed and optimized stored procedures for data transformation and workflow automation, reducing ad-hoc query load on production systems.

Drive Test Engineer

Aug 2015 – Aug 2021

Huawei Technologies Nepal, Lalitpur, Nepal

- Conducted 2G/3G network KPI analysis and cluster drive testing across regional deployments; generated performance reports and resolved customer network issues to ensure SLA compliance.

PROJECTS

• Kansas BWA Data & NLP Pipeline

Built a scalable Python data engineering pipeline orchestrated with Apache Airflow, integrating web scraping, custom spaCy NER, Gemini LLM classification, fuzzy string matching, and geospatial analysis to extract, classify, and map boil water advisory causes from KDHE public health data across Kansas.

• Full-Stack AI RAG Application

Developed a secure, multi-user PDF Q&A platform using FastAPI, Next.js, and LlamaIndex, leveraging Google Gemini for reasoning and Pinecone for vector storage to enable real-time, context-aware document analysis with isolated user sessions.

• Financial Sentiment Analysis (ProsusAI/FinBERT)

Fine-tuned a transformer-based FinBERT model for financial sentiment classification, achieving 87% accuracy and outperforming baseline models including original FinBERT, DistilBERT, LSTM, and Naive Bayes.

• Credit Risk Modeling

Built loan default prediction models using XGBoost, Logistic Regression, and Random Forest; XGBoost achieved up to 99% accuracy across cross-validated evaluation sets.

- **AWS ETL Pipeline for CSV and API Data**
Developed an automated AWS ETL pipeline to extract and validate data from CSV files and REST APIs, event-triggered via AWS SQS.
- **AI-Powered Tableau Dashboard Generator** (*In Progress*)
Building an end-to-end pipeline to analyze datasets, recommend KPIs and visual types, and auto-generate publication-ready Tableau dashboards using LLM reasoning.

TECHNICAL SKILLS

Programming	Python, SQL, TypeScript, Node.js, Next.js, React, Bash/Linux
ML & AI	Scikit-learn, TensorFlow, Keras, PyTorch, CNNs, RNNs, LSTMs, Transformers, Computer Vision, NLP (spaCy, NER)
LLM Frameworks	LangChain, LangGraph, LlamaIndex, OpenAI API, Claude API, Gemini API
Data & ETL	Pandas, NumPy, PySpark, Apache Airflow, Azure Data Factory, AWS, Snowflake, SSIS
Databases	PostgreSQL, MSSQL, MySQL, MongoDB, ChromaDB, Pinecone
APIs & Backend	FastAPI, Flask, REST APIs, GraphQL, SQLAlchemy, Prisma
Visualization	Tableau, PowerBI, Matplotlib, Seaborn, GeoPandas
DevOps & Tools	Docker, Git, GitHub, Jenkins, Selenium, BeautifulSoup

EDUCATION

- **Master of Science, Data Science** **Aug 2024 – May 2026 (Expected)**
Wichita State University, Wichita, Kansas
- **Master of Business Administration** **Aug 2019 – 2023**
DAV Business School, Lalitpur, Nepal
- **Bachelor of Electronics and Communication Engineering** **Aug 2010 – 2014**
Kantipur Engineering College, Lalitpur, Nepal